

💡 Clarifying Stock Index Futures Fundamentals

Before the notes, let's fix the two biggest points of confusion:

1. What is the Futures Price of a Stock Index?

- A stock index (like the S&P 500) is **not a security** you can buy or sell directly; it's just a number.
- A **Stock Index Future** is a contract based on that number. The futures price you see (e.g., 5,000) is the index value that parties agree to settle on in the future.

2. Why is the Price Multiplied by a Multiplier (Contract Size)?

- Since you can't deliver the S&P 500 index itself, the contract value must represent a certain amount of **cash**.
- The **Multiplier** (or Contract Size) converts the index number into a dollar amount. It determines the **value of one contract**.

$$\text{Contract Value}(V_F) = \text{Futures Price} \times \text{Multiplier}$$

- Example (Mini S&P 500):** If the futures price is 5,000 and the multiplier is **\$50**:

$$\text{Contract Value} = 5,000 \times \$50 = \text{\$250,000}$$

When the index moves by **1 point** (e.g., from 5,000 to 5,001), the contract's value changes by **\$50** (1 point $\times$ \$50). This is how profits and losses are calculated.

📄 Chapter 3: Hedging Strategies Using Futures (Part 4)

5. Stock Index Futures (Section 3.5)

A. Understanding the Stock Index

- Definition:** A stock index (like the S&P 500 or Dow Jones) tracks the price changes of a **hypothetical portfolio of stocks**. It is a single number representing the overall movement of a specific market segment.
- Index Calculation:**
 - The index tracks **capital gain/loss** from investing in the portfolio.
 - Dividends are generally excluded** (unless it's a "total return index").
- Index Weighting:** How much influence each stock has on the index value:
 - i. **Price-Weighted:** Weights are proportional to the stock's market **price**. (e.g., **Dow Jones Industrial Average**).
 - ii. **Value-Weighted (Capitalization-Weighted):** Weights are proportional to the stock's **Market Capitalization** (Stock Price $\times$ Shares Outstanding). (e.g., **S&P 500**).

B. Stock Index Futures Contracts

- Key Feature: Cash Settlement:** Futures contracts on stock indices are **always settled in cash** because physically delivering the underlying portfolio (e.g., 500 stocks for the S&P 500) is impossible.
- Settlement:** Contracts are marked to market to the opening or closing price of the index on the last trading day.
 - Example:* S&P 500 futures close out at the opening price of the S&P 500 index on the **third Friday of the delivery month**.
- Contract Multipliers (The "Multiplier Shit"):**

Index/Contract	Multiplier (Contract Size)	Contract Value Example (Index = 5,000)
Mini S&P 500	\$50	$5,000 \times \$50 = \text{\$250,000}$
Mini DJ Industrial Avg	\$5	$5,000 \times \$5 = \text{\$25,000}$
Mini Nasdaq-100	\$20	$5,000 \times \$20 = \text{\$100,000}$

C. Hedging a Well-Diversified Equity Portfolio

Stock index futures are the primary tool for hedging **market risk** (systematic risk) in an equity portfolio.

Case 1: Portfolio Mirrors the Index ($\beta = 1.0$)

If your portfolio is well-diversified and moves exactly like the index, the optimal hedge ratio is 1.0. You hedge the total value of your portfolio using the total value of the futures contracts.

- **Formula for Number of Contracts (N):**

$$N = \frac{V_A}{V_F}$$

Where:

- V_A : Current **Value of the Portfolio** (Exposure).
- V_F : Current **Value of One Futures Contract** (Futures Price $\times$ Multiplier).

- **Example:**

- Portfolio Value (V_A) = **\$5,050,000**
- S&P 500 Futures Price = 1,010
- Contract Multiplier (Value) = \$250
- $V_F = 1,010 \times \$250 = \$252,500$
- $N = \$5,050,000 / \$252,500 = \mathbf{20}$ contracts
- **Action:** Short **20** contracts to hedge the long equity position against a market drop.

Case 2: Portfolio Does Not Mirror the Index ($\beta \neq 1.0$)

When a portfolio is more or less sensitive to market moves, we use **Beta** (β) from the Capital Asset Pricing Model (CAPM) to adjust the hedge ratio.

- **Understanding Beta (β):** β measures the portfolio's volatility relative to the index.
 - $\beta = 1.0$: Portfolio moves exactly like the index.
 - $\beta = 2.0$: Portfolio is **twice** as sensitive (risky) as the index.
 - $\beta = 0.5$: Portfolio is **half** as sensitive as the index.
- **Formula for Hedging (Reducing β to zero):**

$$N = \beta \frac{V_A}{V_F}$$

- *Intuition:* If your portfolio has a $\beta = 2.0$, you must use **twice** the number of contracts (N) to completely offset the market moves.

D. Applications: Changing Portfolio Beta

Futures allow fund managers to adjust the market exposure of their portfolio **without incurring high transaction costs** from selling and buying all the underlying stocks.

- **Goal 1: Complete Hedging ($\beta_T = 0$)**
 - *Action:* Short $N = \beta \frac{V_A}{V_F}$ contracts. The total return will be close to the risk-free rate, regardless of market performance.
- **Goal 2: Changing Beta to a Target (β_T)**
 - **To Reduce Beta ($\beta_T < \beta$):** Short the difference in risk.

$$N = (\beta - \beta_T) \frac{V_A}{V_F}$$

Example: To change β from 1.5 to 0.75, short 15 contracts.

- **To Increase Beta ($\beta_T > \beta$):** Go long (buy) the futures contract to add risk exposure.

$$N = (\beta_T - \beta) \frac{V_A}{V_F}$$

Example: To change β from 1.5 to 2.0, long 10 contracts.

- **Goal 3: Locking in Stock Picking Benefit:** If a manager believes their individual stocks will **outperform** the market, they can short index futures (using $N = \beta \frac{V_A}{V_F}$) to hedge the **market risk** and leave only the potential excess return (alpha) from their stock selection.

Changing Portfolio Beta (β) Using Futures

The Beta (β) of a portfolio measures its **sensitivity to the overall market** (the index).

The general formula for changing the portfolio's current beta (β) to a target beta (β_T) is:

$$N = (\beta - \beta_T) \frac{V_A}{V_F}$$

- N : Number of contracts to trade (short if $\beta > \beta_T$, long if $\beta < \beta_T$).
- V_A : Current value of the portfolio.
- V_F : Current value of one futures contract.

Scenario 1: Complete Market Neutralization ($\beta_T = 0$)

Action	Why Use $\beta_T = 0$?	Formula Result	Position
Hedge Out Market Risk	The portfolio manager is uncertain about the direction of the broad market (index). They want to eliminate systematic risk and isolate the stock-specific risk (alpha).	$N = \beta \frac{V_A}{V_F}$	Short N contracts (assuming $\beta > 0$).

Detailed Explanation

When you set the target beta (β_T) to zero, you are achieving a **Market Neutral** position.

- **Goal:** To earn a return equal to the risk-free rate plus the "alpha" (excess return) generated by the manager's individual stock selection. The performance of the portfolio then depends *only* on the stocks chosen, not on whether the entire market goes up or down.
- **When to Use:**
 - **Market Fear:** The manager expects a massive market downturn but does not want to sell their carefully chosen stocks due to high transaction costs.
 - **Focus on Alpha:** The manager wants to clearly demonstrate their stock-picking ability. If the overall market falls, but the hedged portfolio still earns a positive return, that return is pure alpha.
- **Example:** A portfolio has $\beta = 1.2$. The manager shorts 1.2 times the market value in futures. If the market drops 10%, the portfolio should lose 12%. But the futures position gains exactly enough cash to offset that 12% loss. The portfolio is protected.

Scenario 2: Decreasing Beta ($\beta_T < \beta$)

Action	Why Use $\beta_T < \beta$?	Formula Result	Position
Reduce Market Risk Exposure	The portfolio manager believes the market will face a moderate decline or become highly volatile, but does not want to completely eliminate all market exposure.	$N = (\beta - \beta_T) \frac{V_A}{V_F}$	Short N contracts.

Detailed Explanation

This strategy is used to **de-risk** the portfolio without fully neutralizing it.

- **Goal:** To reduce the portfolio's sensitivity to market swings, keeping some exposure to market gains while reducing the impact of market losses.
- **When to Use:**
 - **Cautious Outlook:** The manager forecasts market volatility but isn't bearish enough to go to $\beta = 0$. By lowering β from 1.0 to 0.5, the portfolio only moves half as much as the market.
 - **Example:** A portfolio currently has $\beta = 1.5$. The manager wants to reduce it to $\beta_T = 0.75$.

- They calculate the required number of short contracts based on the difference: $N \propto (1.5 - 0.75) = 0.75$. They short enough contracts to hedge away 0.75 of their current market exposure.

Scenario 3: Increasing Beta ($\beta_T > \beta$)

Action	Why Use $\beta_T > \beta$?	Formula Result	Position
Increase Market Risk Exposure	The portfolio manager is extremely bullish on the market but does not want to spend the cash or time required to buy more of the underlying stocks.	$N = (\beta_T - \beta) \frac{V_A}{V_F}$	Long N contracts.

Detailed Explanation

This strategy is used to **lever up** the portfolio's market exposure easily and cheaply.

- **Goal:** To gain extra exposure to the market's expected rise using a leveraged instrument (futures), which only requires a small margin deposit.
- **When to Use:**
 - **Very Bullish Outlook:** The manager expects a strong rally and wants the portfolio's return to exceed the market return.
 - **Capital Constraints:** The manager has excess cash available for investment but wants to deploy it slowly. By buying index futures, they gain immediate market exposure without fully committing capital to buying stocks.
 - **Example:** A portfolio currently has $\beta = 1.0$. The manager wants to increase it to $\beta_T = 1.8$.
 - They calculate the required number of long contracts based on the difference: $N \propto (1.8 - 1.0) = 0.8$. They long enough contracts to add 0.8 of market exposure to their existing 1.0.

Summary Table of Portfolio Beta Strategy

Target Beta (β_T) vs. Current Beta (β)	Strategic Goal	Futures Position Required	Effect on Portfolio Risk
$\beta_T = 0$	Eliminate systematic risk; capture pure alpha.	Short contracts (β amount).	Portfolio return becomes independent of the market.
$\beta_T < \beta$	Reduce systematic risk; moderate bearish outlook.	Short contracts ($\beta - \beta_T$ amount).	Portfolio risk and returns are moderated.
$\beta_T > \beta$	Increase systematic risk; bullish outlook.	Long contracts ($\beta_T - \beta$ amount).	Portfolio risk and returns are amplified (leveraged).

6. Stack and Roll (Section 3.6)

A. The Challenge and The Strategy

- **The Challenge:** The required hedge period (T) is **longer** than the maturity of the most liquid futures contracts available.
- **Stack and Roll:** The strategy of **rolling the hedge forward** by sequentially closing out the near-term contract and immediately opening the same position in the next contract.
- **Procedure (Short Hedge Example):**
 - Time t_1 :** Short Contract 1 (e.g., October).
 - Time t_2 (September):** Close out Short October contract for a profit/loss. Immediately **Short Contract 2** (e.g., March).
 - Time t_3 (February):** Close out Short March contract. Immediately **Short Contract 3** (e.g., July).
 - Time T (June):** Close out final contract and sell the asset spot.

B. Risk of Stack and Roll

- **Basis Risk is Amplified:** Each time you roll the hedge, you incur **new basis risk** between the two successive futures contracts (e.g., the difference between the October and March futures prices). This uncertainty makes the final locked-in price less predictable.

C. Liquidity Risk and Cash Flow Mismatch

- **Metallgesellschaft (MG) Disaster:** MG used a stack and roll strategy (long futures) to hedge long-term fixed-price sales contracts.
- **The Mismatch:** Oil prices unexpectedly **fell**.
 - **Futures Side (Hedge):** MG's long futures position lost money daily, triggering massive **margin calls** (immediate cash outflow).
 - **Sales Contract Side (Exposure):** The value of their long-term fixed-price contracts increased (expected long-term gain).
- **Result:** The huge, immediate cash drain from the daily margin calls caused severe **liquidity problems** that the company could not manage, forcing them to close the hedge at a massive loss (\$1.33 billion).
- **Moral:** A hedging strategy must not only manage price risk but also ensure the company has sufficient cash flow to meet **margin calls** that result from temporary adverse price movements.